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PHYSICS

1) A series LCR circuit containing 8.0 H inductor, 50 μF capacitor and 80 Ω resistor is connected to 220 V variable frequency ac source. The angular frequencies of the source at which power transferred to the circuit is half the power at resonant angular frequency are likely to be :

(1) 25 rad/s and 75 rad/s

(2) 30 rad/s and 50 rad/s

(3) 40 rad/s and 60 rad/s

(4) 45 rad/s and 55 rad/s

2) Consider the nuclear reaction $X^{200} \rightarrow A^{120} + B^{80}$. If the binding energy per nucleon for X, A and B are 7.4 MeV, 8.2 MeV and 8.2 MeV respectively, then the energy released in the reaction is :-

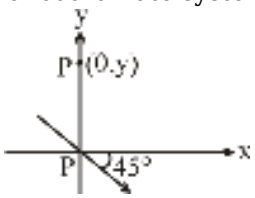
(1) 160 MeV

(2) 200 MeV

(3) 190 MeV

(4) 10 MeV

3) A short electric dipole is placed at origin of coordinate system as shown in figure. Find the electric field at point P(0, y). $\left(k = \frac{1}{4\pi\in_0}\right)$:



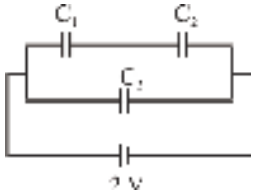
(1) $\left(\sqrt{\frac{5}{2}}\right)\frac{kP}{y^3}$

(2) $\sqrt{5}\frac{kP}{y^3}$

(3) $\left(\sqrt{\frac{3}{2}}\right)\frac{kP}{y^3}$

(4) $\sqrt{3}\frac{kP}{y^3}$

4) Two capacitors $C_1 = 2$ μF and $C_2 = 6$ μF in series, are connected in parallel to a third capacitor $C_3 = 4$ μF. This arrangement is then connected to a battery of e.m.f. = 2 V, as shown in the figure. How much energy is given by the battery in charging the capacitors ?



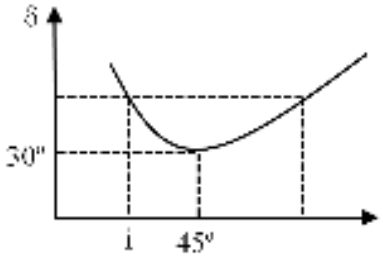
(1) 22×10^{-6} J

(2) 11×10^{-6} J

(3) $\left(\frac{32}{3}\right) \times 10^{-6}$ J

(4) $\left(\frac{16}{3}\right) \times 10^{-6}$ J

5) A graph between angle of incidence and angle of deviation for a prism placed in air is as shown in figure. The refractive index and critical angle for prism, for given pair of medium are respectively :-



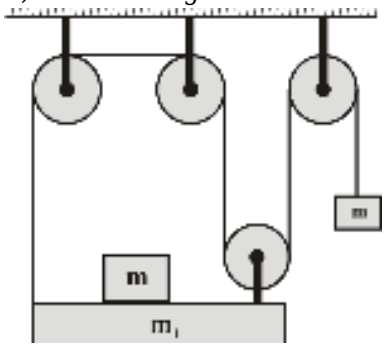
(1) 1.5, 30°

(2) 1.25, 37°

(3) $\sqrt{2}$, 45°

(4) $\sqrt{2}$, 60°

6) If the arrangement shown in the figure is in equilibrium, then mass of m_1 will be equal to :-



(1) m/2

(2) m

(3) 2m

(4) zero

7) The orbital velocity of an artificial satellite in a circular orbit very close to earth is V. The velocity of a geo-sationary satellite orbiting in circular orbit at an altitude of 3R from earth's surface will be :

(1) $\frac{V}{\sqrt{7}}$

(2) $\frac{V}{\sqrt{6}}$

(3) $\frac{V}{2}$

(4) $\frac{V}{\sqrt{2}}$

8) Match the following columns and mark the correct option from the codes given below.

	Column I		**Column II**
(A)	Constant positive acceleration	(p)	Speed may increase
(B)	Constant negative acceleration	(q)	Speed may decrease
(C)	Constant position	(r)	Speed is zero
(D)	Constant slope of v-t graph	(s)	Speed must increase
		(t)	Speed must decrease

Codes

	A	**B**	**C**	**D**
(1)	q, r	s	p	t
(2)	p, q	p, q	r	p, q
(3)	p	s, r	s	q
(4)	q	p	t	s, q

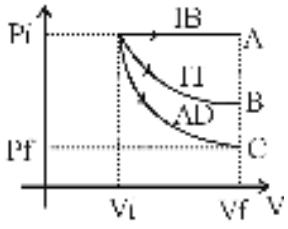
(1) 1

(2) 2

(3) 3

(4) 4

9) Which is correct ?



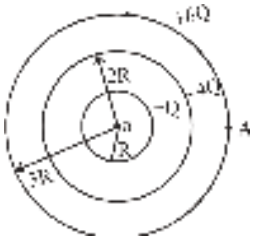
(IB→Isobaric; IT→Isothermal; AD→Adiabatic)

(1) $U_A > U_B$

(2) $U_B > U_C$

(3) $U_A > U_C$

(4) All

10)  There are three concentric conducting spheres of radius R, 2R, 3R having charge +Q,–4Q, + 6Q respectively. Then find magnitude of electric field at point 'A'.

(1) $\frac{Q}{4\pi\varepsilon_0 R^2}$

(2) $\frac{Q}{9\pi\varepsilon_0 R^2}$

(3) $\frac{Q}{12\pi\varepsilon_0 R^2}$

(4) $\frac{Q}{16\pi\varepsilon_0 R^2}$

11) A dog weighing 5 kg is standing on a flat boat so that dog is 10 m from the shore. The dog walks 4 m on the boat towards the shore and then halts. The boat weighs 20 kg and one can assume that there is no friction between it and the water. How far is the dog from the shore at the end of this time?

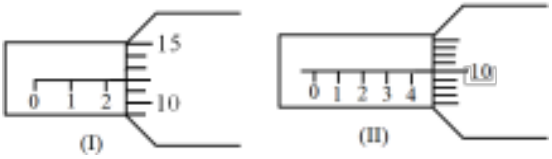
(1) 3.2 m

(2) 0.8 m

(3) 10 m

(4) 6.8 m

12) A screw gauge has some zero error but its value is unknown. There are two identical thin sheets. When the first sheet is placed for thickness measurement, the reading of instrument is shown in figure (I). When both sheets placed together for measurement, with their faces in contact, the reading is as shown in figure (II). The zero error in instrument is : (Assume one full rotation of screw is equivalent to 1 mm and circular scale has 100 divisions.)



(1) –0.16 mm

(2) +0.16 mm

(3) +0.14 mm

(4) –0.14 mm

13) In a simple microscope, image is formed at a distance of 50 cm from the lens. If the focal length of lens is 5 cm, then find out its magnifying power.

(1) 5

(2) 5.5

(3) 0.5

(4) 4.5

14) A resonance tube length of 1.00 m. If speed of sound at room temperature is 340 ms^{-1}, then neglecting end correction, the minimum frequency of tuning fork so that at least three resonance can occur in it, is :

(1) 400 Hz

(2) 425 Hz

(3) 475 Hz

(4) 525 Hz

15) One mole of an ideal gas at temperature T_1 expands according to the law $\frac{P}{V^2}$ = a (constant). The work done by the gas till temperature of gas becomes T_2 is :-

(1) $\frac{1}{2}R(T_2 - T_1)$

(2) $\frac{1}{3}R(T_2 - T_1)$

(3) $\frac{1}{4}R(T_2 - T_1)$

(4) $\frac{1}{5}R(T_2 - T_1)$

16) A metal block of mass 100g at 120°C is dropped into a mixture of 80g ice and 100 g water inside a calorimeter of water equivalent 20g. The amount of ice that melts when equilibrium is reached is : (Given : Specific heat of metal = 0.10 $calg^{-1}°C^{-1}$; Latent heat of fusion of ice = 80 cal g^{-1})

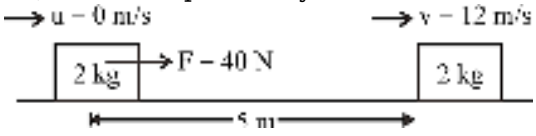
(1) 8g

(2) 11g

(3) 15g

(4) 21g

17) A block is pushed by force of 40 N as shown in diagram then work done by friction will be :-



(1) –200 J

(2) – 300 J

(3) – 44 J

(4) – 56 J

18) A certain metallic surface is illuminated with monochromatic light of wavelength, λ. The stopping potential for photo-electric current for this light is $3V_0$. If the same surface is illuminated with light of wavelength 2λ, the stopping potential is V_0. The threshold wavelength for this surface

for photoelectric effect is :-

(1) 4 λ

(2) $\frac{\lambda}{4}$

(3) $\frac{\lambda}{6}$

(4) 6 λ

19) A charge '+q' moving with a velocity $\vec{V} = V_0\hat{i}$ entres a region of uniform electric and magnetic fields with $\vec{E} = 4 \times 10^3\hat{j}$ N/C and $\vec{B} = 0.1\hat{k}T$. As it moves in the two fields, it deflects along negative y-axis. Which of the following could be possible value of V_0 ?

(1) 3×10^4 m/s

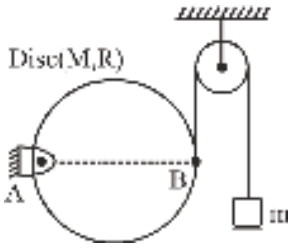
(2) 4×10^4 m/s

(3) 5×10^4 m/s

(4) 2×10^4 m/s

20)

Find the value of mass m to keep the disc horizontal. (Given string is attached at point B on disc.)



(1) M

(2) $\frac{M}{2}$

(3) $\frac{M}{4}$

(4) $\frac{M}{3}$

21) A circular platform of moment of inertia 100 kg m^2 with a boy of mass 20 kg standing at the centre, completes one rotation in 10s. If he walks 2m away from centre along radius, then one rotation (with respect to ground) will be completed in :-

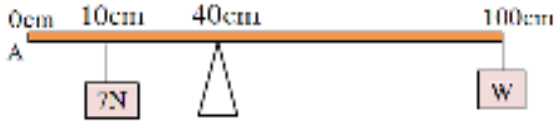
(1) 12s

(2) 18s

(3) 16s

(4) 20s

22) A uniform meter scale weighing 3.0N is to be balanced on knife edge placed under the 40cm mark and carrying a weight of 7.0N hanging from 10cm mark. The scale is kept balanced horizontally by a weight W hanging from the 100cm end. Calculate the value of weight W.



(1) 2N

(2) 4N

(3) 3N

(4) 5N

23) A body is thrown up with a speed 49 m/s. It travels 5m in the last second of its ascent journey. If the same body is thrown up with a velocity 98 m/s, how much distance (in m) will it travel in the last second of its ascent ?
($g = 10m/s^2$)

(1) 10 m

(2) 49 m

(3) 5 m

(4) 15 m

24) To find the resistance of galvanometer by half deflection method, experimental data obtained is given in table below :

S. No.	Resistance R (Ω)	Deflection in galvanometer (θ)	Shunt resistance S (Ω)	Half deflection (θ/2)	Galvanometer resistance (G)
1	3300	30	80	15	G_1
2	5000	20	80	10	G_2

From the above data, the galvanometer resistance R will be near to :

(1) 20Ω

(2) 40Ω

(3) 50Ω

(4) 80Ω

25) A steel wire of cross-sectional area $3\times10^{-6}m^2$ can withstand a maximum strain of 10^{-3}. Young's modulus of steel is 2×10^{11} N/m^2. The maximum mass the wire can hold is (Take $g = 10m/s^2$) :-

(1) 40kg

(2) 60kg

(3) 80kg

(4) 100kg

26) The velocity of water waves v may depend upon their wavelength λ, the density of water ρ and the acceleration due to gravity g . The method of dimensions gives the relation between these quantities as (here λ is measured in terms of length)

(1) $v^2 \propto \lambda g^{-1} \rho^{-1}$

(2) $v^2 \propto g\lambda\rho$

(3) $v^2 \propto g\lambda$

(4) $v^2 \propto g^{-1}\lambda^{-3}$

27) Two beams, A and B, of plane polarized light with mutually perpendicular planes of polarization are seen through a polaroid. From the position when the beam A has maximum intensity (and beam B has zero intensity), a rotation of polaroid through 30º makes the two beams appear equally bright. If the initial intensities of the two beams are I_A and I_B respectively, then $\frac{I_A}{I_B}$ equals to :-

(1) 3

(2) 3/2

(3) 1

(4) 1/3

28) A hole is there in the bottom of an open tank having water. If total pressure at bottom is 4 atm (1 atm = 10^5 Pa), then velocity of water flowing from hole (in m/s) is :-

(1) $\sqrt{400}$

(2) $\sqrt{600}$

(3) $\sqrt{100}$

(4) $\sqrt{300}$

29) Choose the essential conditions to obtain a sharp null point in meter bridge experiment :
a. The bridge wire should have uniform cross section.
b. The resistance in two gaps should be of nearly same magnitude.
c. The galvanometer should have high resistance.
d. Using thicker wire for bridge wire

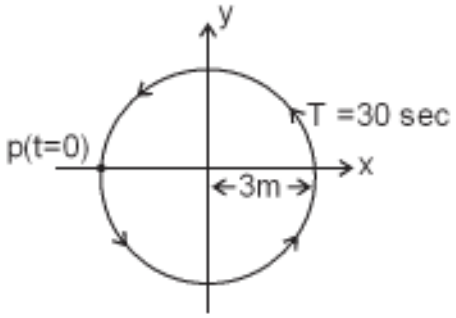
(1) a & d

(2) a & c

(3) b & c

(4) a & b

30) For a particle p the radius of the circle, the period of revolution, initial position and direction of revolution are indicated in the figure.



The y-projection of the radius vector of rotating particle p will be-

(1) $y = 3\sin\left(\frac{\pi}{15}t\right)$

(2) $y = -3\sin\left(\frac{\pi}{15}t\right)$

(3) $y = 3\cos\left(\frac{\pi}{15}t\right)$

(4) $y = -3\cos\left(\frac{\pi}{15}t\right)$

31) The resistance of galvanometer coil is 0.1 KΩ. The current for full scale deflection is 100μA. The value of the resistance put in series to convert it into a voltmeter of range 0.1 volt is :-

(1) 1000 Ω

(2) 100 Ω

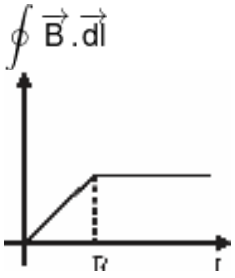
(3) 10 Ω

(4) 900 Ω

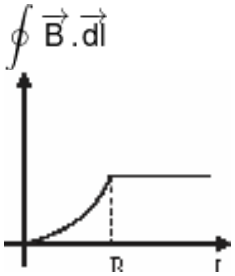
32) A cylindrical wire of a radius R is carrying current i uniformly distributed over its cross section.

If a circular loop of radius 'r' is taken as amperian loop, then the variation of value of $\oint \vec{B}.\vec{dl}$ over this loop with radius 'r' of loop will be best represented by

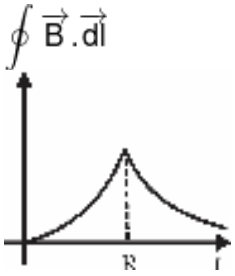
(1)

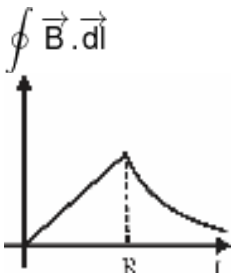


(2)



(3)



(4) 

33) Ratio of longest wavelengths corresponding to Lyman and Balmer series in hydrogen spectrum is:-

(1) $\frac{9}{31}$

(2) $\frac{5}{27}$

(3) $\frac{3}{23}$

(4) $\frac{7}{29}$

34) A metal ball of surface area 200 cm^2 and temperature 527°C is surrounded by a vessel at 27°C. If the emissivity of the metal is 0.4, then the rate of loss of heat from the ball is :-
($\sigma = 5.67 \times 10^{-8}$ J / m^2 – s – K^4)

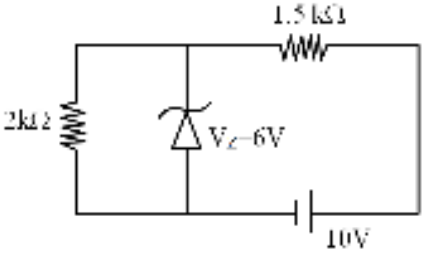
(1) 108 joule/sec (approx)

(2) 168 joule/sec (approx)

(3) 182 joule/sec (approx)

(4) 192 joule/sec (approx)

35) For the circuit shown in figure, the current flowing through 2kΩ resistor is -



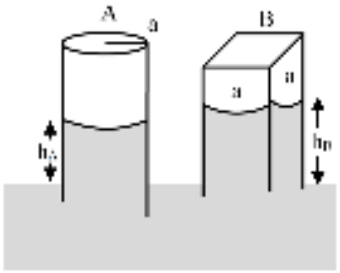
(1) 2.5 mA

(2) 2.85 mA

(3) 3 mA

(4) 4.5 mA

36) Two capillary tubes A and B of same length and material are partially dipped inside a liquid of density ρ. Capillary A has circular cross-section of radius 'a'. While capillary B has square cross



section of side a. The ratio of rise of liquid in capillary A and B is :

(1) π : 1

(2) 1 : π

(3) 1 : 2

(4) 2 : 1

37) The fundamental frequency of a closed pipe is 220 Hz. If $\frac{1}{4}$ of the pipe is filled with water, the frequency of the first overtone of the pipe now is :

(1) 220 Hz

(2) 440 Hz

(3) 880 Hz

(4) 1760 Hz

38) **Assertion :** A photo diode is used in R.B. detect the intensity of light.
Reason : In R.B condition current is small but it is more sensitive to changes in incident light intensity.

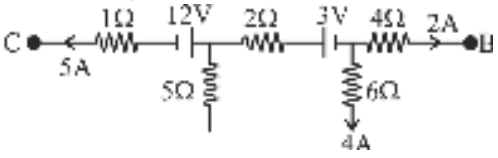
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

39) The figure shows part of certain circuit, find potential difference $V_C - V_B$:-



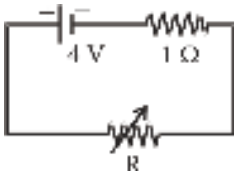
(1) 6 V

(2) –6 V

(3) 8 V

(4) 10 V

40) For different values of resistance R, power consumptions in R are given. Then which of the



following values are not possible ? (a) 2W (b) 5W
(c) 8 W (d) 4W

(1) Only c
(2) b & c
(3) a,b,c
(4) All

41) A bulb is placed at a depth of $2\sqrt{7}$ m in water and a floating opaque disc is placed over the bulb so that the bulb is not visible from the surface. What is the minimum diameter of the disc? ($\mu_w = 4/3$)

(1) 6
(2) 3
(3) 5
(4) 12

42) If $|\vec{A}| = 2$ and $|\vec{B}| = 4$ and angle between them is 60°, then $\left|\vec{A}-\vec{B}\right|$ is :-

(1) $\sqrt{13}$
(2) $3\sqrt{3}$
(3) $\sqrt{3}$
(4) $2\sqrt{3}$

43) The ratio of de-Broglie wavelengths of molecule of hydrogen and helium atom, which are at temperature 27°C and 127°C respectively, is

(1) $\frac{1}{2}$
(2) $\sqrt{\frac{3}{8}}$
(3) $\sqrt{\frac{8}{3}}$
(4) 1

44) An EMW is propagating in free space, then total average energy density is given by (E_0, B_0 are peak values) :-

(1) $\frac{1}{4}\varepsilon_0 E_0^2$
(2) $\frac{1}{2}\varepsilon_0 E_0^2 + \frac{1}{2}\frac{B_0^2}{\mu_0}$

(3) $\frac{1}{4}\varepsilon_0 E_0^2 + \frac{1}{4}\frac{B_0^2}{\mu_0}$

(4) $\frac{1}{4}\frac{B_0^2}{\mu_0}$

45) A solenoid of self inductance 1 mH have the length 1m, then find out the length of wire by which the solenoid is constructed :-

(1) 10 m.

(2) 100 m.

(3) 1000 m.

(4) 104 m.

CHEMISTRY

1) For the hypothetical reaction

$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$

$\Delta_r G°$ and $\Delta_r S°$ are 20 kJ/mol and – 20 JK^{-1} mol^{-1} respectively at 200 K. If $\Delta_r C_p$ is 20 $JK^{-1}mol^{-1}$ then $\Delta_r H°$ at 400 K is :-

(1) 20 kJ/mol

(2) 7.98 kJ/mol

(3) 28 kJ/mol

(4) None of these

2) Aqueous solution contains 5% urea and 10% glucose (by weight). Its freezing point will be (K_f = 1.86°Cm^{-1}) :-

(1) –2.04 °C

(2) –3.04 °C

(3) –1.04 °C

(4) None

3) The most important buffer in the blood consists of :

(1) HCl and Cl^-

(2) H_2CO_3 and HCO_3^-

(3) H_2CO_3 and Cl^-

(4) HCl and HCO_3^-

4) Which electron is associated with the least energy?

(1) n = 3, □ = 2, m = –2, $s = +\frac{1}{2}$

(2) n = 4, ℓ = 0, m = 0, $s = +\frac{1}{2}$

(3) n = 4, ℓ = 1, m = +1, $s = +\frac{1}{2}$

(4) n = 5, ℓ = 0, m = 0, $s = +\frac{1}{2}$

5) The orbital angular momentum of electron in f subshell will be :-

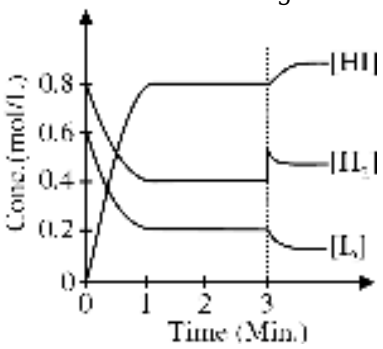
(1) $\sqrt{6}\hbar$

(2) $\sqrt{20}\hbar$

(3) $\sqrt{2}\hbar$

(4) $\sqrt{12}\hbar$

6) Variation of concentration with time for following reaction in the figure below is:



$H_2(g) + I_2(g)$ – heat ⇌ 2HI(g) At the instant 3 min, what change was imposed into the equilibrium ?

(1) Pressure was increased

(2) Temperature was decreased

(3) Temperature was increased

(4) Hydrogen was added

7) Which of the following is incorrect relation?

(1) 1 µg/mL = 1 ppm

(2) 1 mg/L = 1 ppm

(3) 1 g/L = 1 ppm

(4) 1 g/kL = 1 ppm

8) Correct expression for the first order reaction :-

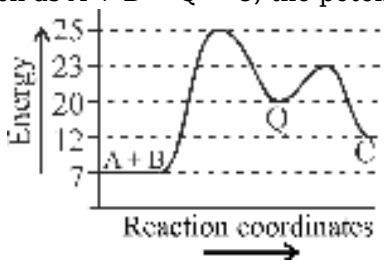
(1) $C_t = C_0 e^{k_1 t}$

(2) $C_t e^{k_1 t} = C_0$

(3) $\ell n \left(\frac{C_0}{C_t}\right) = -k_1 t$

(4) $\ell n \frac{C_t}{C_0} = k_1 t$

9) For a multistep reaction such as A + B → Q → C; the potential energy diagram is shown below.



What is E_a for the step Q → C?

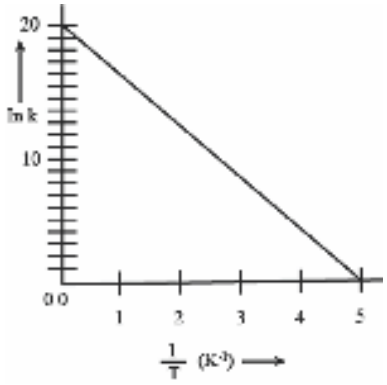
(1) 3 kcal mol^{-1}

(2) 5 kcal mol^{-1}

(3) 8 kcal mol^{-1}

(4) 11 kcal mol^{-1}

10) For a reaction, given below is the graph of (ln k vs $\frac{1}{T}$). The activation energy for the reaction is equal to.....$calmol^{-1}$.



(Given: R = 2$calK^{-1}mol^{-1}$)

(1) 8

(2) 4

(3) 20

(4) 5

11) MnO_4^{2-} (1 mole) in neutral aqueous medium is disproportionate to :

(1) 2/3 mole of MnO_4^-, 1/3 mole of MnO_2

(2) 1/3 mole of MnO_4^-, 2/3 mole of MnO_2

(3) 1/3 mole of Mn_2O_7, 1/3 mole of MnO_2

(4) 2/3 mole of Mn_2O_7, 1/3 mole of MnO_2

12) The value of x in the following reaction

$MnO_4^- + 8H^+ + xe^- \rightleftharpoons Mn^{+2} + 4H_2O$ is :-

(1) 5

(2) 3

(3) 1

(4) 0

13) If ΔH_f° of $ICl_{(g)}$, $Cl_{(g)}$, and $I_{(g)}$ is 17.57, 121.34 and 106.96 J mol^{-1} respectively. Then bond dissociation energy of ICl bond is :-

(1) 35.15 J mol^{-1}

(2) 106.69 J mol^{-1}

(3) 210.73 J mol^{-1}

(4) 420.9 J mol^{-1}

14) Match the column :-

	Column-I		Column-II
(i)	$NaCl_{(s)} \rightarrow Na^+_{(aq)} + Cl^-_{(aq)}$	(p)	Enthalpy of atomisation
(ii)	$NaCl_{(s)} \rightarrow Na^+_{(g)} + Cl^-_{(g)}$	(q)	$\Delta_{sol}H^0$
(iii)	$Na_{(s)} \rightarrow Na_{(g)}$	(r)	Enthalpy of formation
(iv)	$H_{2(g)} + \frac{1}{2} O_{2(g)} \rightarrow H_2O_{(\square)}$	(s)	Lattice enthalpy

(1) (i)-q ; (ii)-r ; (iii)-p, (iv)-q

(2) (i)-q ; (ii)-s ; (iii)-r, (iv)-p

(3) (i)-r ; (ii)-s ; (iii)-p, (iv)-q

(4) (i)-q ; (ii)-s ; (iii)-p, (iv)-r

15) If 0.5 mol of $CaBr_2$ is mixed with 0.2 mol of K_3PO_4 then the maximum number of moles of $Ca_3(PO_4)_2$ obtained will be :-
$3CaBr_2 + 2K_3PO_4 \rightarrow Ca_3(PO_4)_2 + 6KBr$

(1) 0.5

(2) 0.2

(3) 0.7

(4) 0.1

16) Which of the following has pπ–dπ bonding ?

(1) NO_3^-

(2) SO_3^{2-}

(3) BO_3^{3-}

(4) CO_3^{2-}

17) Which of the following order is incorrect?

(1) Ionic character = $MCl < MCl_2 < MCl_3$

(2) Polarizibility = $Cl^- < Br^- < I^-$

(3) Polarising power = $Na^+ < Ca^{+2} < Mg^{+2} < Al^{+3}$

(4) Covalent character = LiF < LiCl < LiBr < LiI

18) Match **List-I** with **List-II**

List-I	List-II
A. XeF_4	I. See – saw
B.SF_4	II. Square planar
C.NH_4^+	III. Bent T–shaped
D.BrF_3	IV. Tetrahedral

Choose the correct answer from the options given below :

(1) A-IV, B-III, C-II, D-I

(2) A-II, B-I, C-III, D-IV

(3) A-IV, B-I, C-II, D-III

(4) A-II, B-I, C-IV, D-III

19)

Which of the following complex compound shows least C–O bond length?

(1) $[Mn(CO)_6]Cl$

(2) $H[V(CO)_6]$

(3) $[Cr(CO)_6]$

(4) $K_2[Ti(CO)_6]$

20) Which of the following is not optically active ?

(1) $[Co(en)_3]^{3+}$

(2) $[Cr(ox)_3]^{3-}$

(3) cis-$[CoCl_2(en)_2]^+$

(4) trans-$[CoCl_2(en)_2]^+$

21)

	Column -I		Column -II (no. of unpair electron
(A)	$[CrF_6]^{4-}$	(i)	1
(B)	$[MnF_6]^{4-}$	(ii)	2
(C)	$[Cr(CN)_6]^{4-}$	(iii)	4
(D)	$[Mn(CN)_6]^{4-}$	(iv)	5

(1) (A) - iii; (B) - iv; (C) - ii; (D) -i

(2) (A) - iv; (B) - iii; (C) - ii; (D) -i

(3) (A) - i; (B) - ii; (C) - iii; (D) -iv

(4) (A) - i; (B) - ii; (C) - iv; (D) -iii

22) Which of the following statement is **correct** when a mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4?
I. a deep red vapor is evolved.
II. the vapour when passed into NaOH solution gives a yellow solution of Na_2CrO_4
III. Chlorine gas is evolved
IV. chromyl chloride is formed

(1) I, II, IV

(2) I, II, III

(3) II, III, IV

(4) all are correct

23) When 1 mol $CrCl_3.6H_2O$ is treated with excess of $AgNO_3$, 3 mol of AgCl are obtained. The formula of the complex :

(1) $[CrCl_3(H_2O)_3].3H_2O$

(2) $[CrCl_2(H_2O)_4]Cl.2H_2O$

(3) $[CrCl(H_2O)_5]Cl_2.H_2O$

(4) $[Cr(H_2O)_6]Cl_3$

24) Select the correct statement(s) about actinoids among the following.
(a) Neptunium shows highest number of possible oxidation state.
(b) 5f-orbitals present in these elements are much more buried than 4f orbitals of lanthanoids.
(c) Due to comparable energies of 5f, 6d and 7s orbitals, they have smaller range of oxidation states.
(d) +3 and +4 ions of these elements do not hydrolyse.

(1) Only a

(2) Only a and d

(3) Only a and c

(4) a, b, c and d

25) Hybrid state of Aluminium in acidified aqueons solution of $AlCl_3$ is :-

(1) sp^3

(2) sp^3d

(3) sp^3d^2

(4) sp^3d^3

26) Which of the following statements is **not correct** ?

(1) $(CH_3)_3$ COH is less acidic than $(CH_3)_3$ SiOH

(2) Like CO, its analogue of Si is not stable

(3) Bond length [B – F] $BF_3 < BF_4^-$

(4) None of the above is incorrect

27) Which of the following pair of chalcogen group have minimum and maximum electron affinity respectively :-

(1) O, S

(2) S, Po

(3) Po, O

(4) Se, O

28) Give the correct order of initials T or F for following statements. Use T if statement is true and F if it is false :
(i) Cu^+ undergoes disproportionation of Cu and Cu^{2+} in aqueous solution.
(ii) H_2S does not impart sodium Nitro Prusside Test.
(iii) Sulphide ions react with sodium nitroprusside to form a purple coloured complex. In this reaction, oxidation state of iron changes.

(1) TFT

(2) TTF

(3) TFF

(4) FTT

29) The metal that does not give the borax bead test is :

(1) Cr

(2) Ni

(3) Pb

(4) Mn

30) If HCl is not added before passing H_2S in the second group, it may result in the :

(1) incomplete precipitation of second group sulphide

(2) precipitation of sulphides of cations belonging to subsequent groups

(3) precipitation of sulphur

(4) precipitation of lead as lead sulphide

31) In the reaction :

$$Me_3COCH_2CH_3 + \underset{(1\ Mole)}{HI} \xrightarrow{\Delta}$$ Product is :-

(1) $Me_3C - OH + CH_3CH_2–I$

(2) $Me_3C - I + CH_3CH_2OH$

(3) $Me_3C - I + CH_3CH_2I$

(4) $Me_3C - OH + CH_3CH_2OH$

32) Consider the following reaction

$C_2H_2 \xrightarrow[873\ K]{\text{Red hot iron tube}} A$

Incorrect statement about product A is

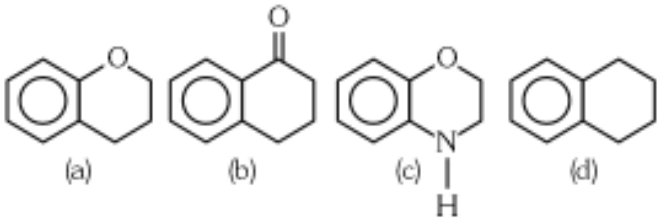
(1) It is an aromatic compound

(2) It reacts with acetic anhydride in presence of anhydrous $AlCl_3$ to give acetophenone

(3) It can be obtained by heating phenol with zinc dust

(4) It decolourises Br_2 water

33) Rank the following compounds in decreasing order of reactivity towards electrophilic aromatic



substitution reaction (ESR).

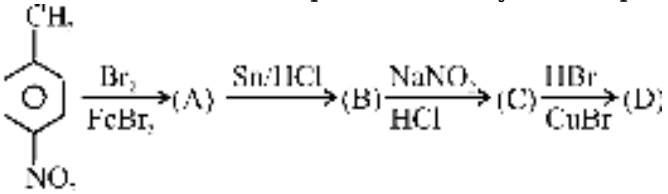
(1) (c) > (a) > (b) > (d)

(2) (c) > (a) > (d) > (b)

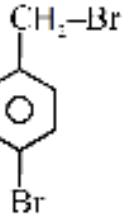
(3) (d) > (c) > (b) > (a)

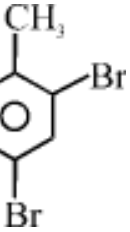
(4) (a) > (c) > (d) > (b)

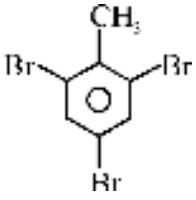
34) In a set of reactions p-nitrotoluene yielded a product (D)

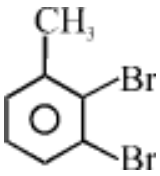


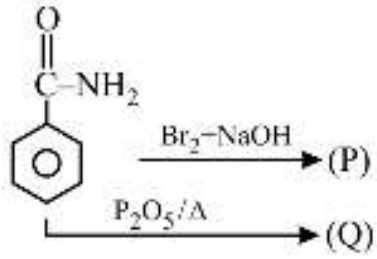
The product (D) would be :-

(1) 

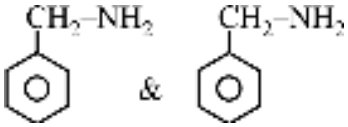
(2) 

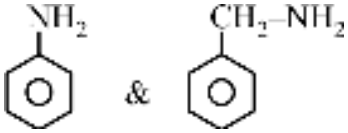
(3) 

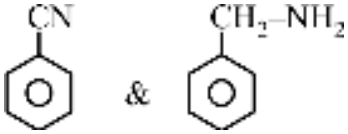
(4) 

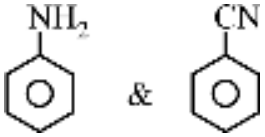
35) 

What are product 'P' & 'Q' respectively.

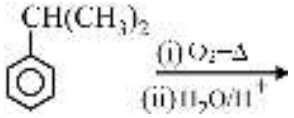
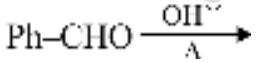
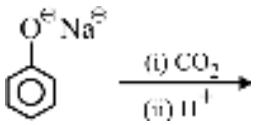
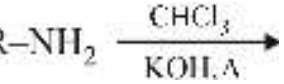
(1) 

(2) 

(3) 

(4) 

36) Match the column.

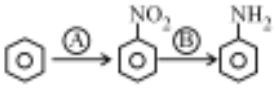
Column - I		Column - II	
(I)		P	Cannizarro reaction
(II)		Q	Kolbe Schmidt reaction
(III)		R	Carbyl amine reaction
(IV)		S	Cumene hydroperoxide method

(1) I → P, II → Q, III → R, IV → S

(2) I → S, II → P, III → Q, IV → R

(3) I → S, II → R, III → Q, IV → P

(4) I → P, II → S, III → R, IV → Q

37) 

A & B Respectively

(1) Sn. HCl, HNO_3. H_2SO_4

(2) HNO_3. H_2SO_4, Sn + HCl

(3) H_2SO_4 , Sn. HCl

(4) $NaBH_4$, HNO_3. H_2SO_4

38) Vitamin C is chemically :-

(1) Ascorbic Acid

(2) Citric Acid

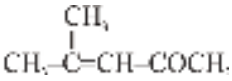
(3) Aspirin

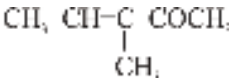
(4) Aspartic Acid

39) $CH_3–CO–CH_3 \xrightarrow[\Delta]{Ba(OH)_2}$ Product will be :-

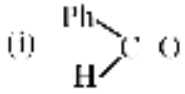
(1) $CH_3–CH=CH–CHO$

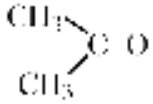
(2) $CH_3–CH_2–CH_3$

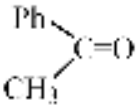
(3) 

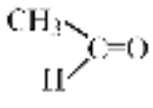
(4) 

40) Give decreasing order of reactivity for nucleophilic addition reaction :-

(i) 

(ii) 

(iii) 

(iv) 

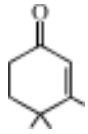
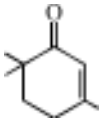
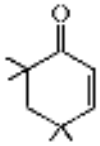
(1) (i) > (ii) > (iii) > (iv)

(2) (iv) > (i) > (ii) > (iii)

(3) (iv) > (i) > (iii) > (ii)

(4) (ii) > (i) > (iv) > (iii)

41) Which of following compounds will show tautomerism :-

(a)  (b)  (c) 

(1) a,b

(2) a,c

(3) b,c

(4) a,b,c

42)

The correct order of acidic strength is :-

(1) $CH_3COOH>CH\equiv CH>HCN> H_2O > C_2H_5OH$

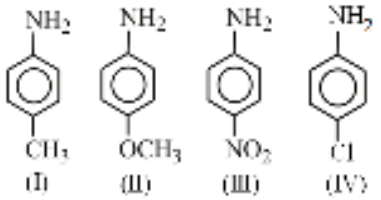
(2) $CH_3COOH>HCN>H_2O> C_2H_5OH > CH\equiv CH$

(3) $HCN>CH_3COOH>CH\equiv CH> C_2H_5OH > H_2O$

(4) $HCN>CH_3COOH>H_2O> C_2H_5OH > CH\equiv CH$

43)

The basic character of substituted anilines (I to IV) are such that



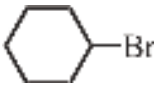
(1) I < II < IV < III

(2) I < II < III < IV

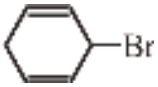
(3) II < I < III < IV

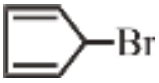
(4) III < IV < I < II

44) The compound which under goes S_{N^1} reaction most rapidly is :-

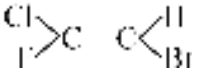
(1) 

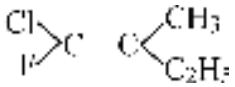
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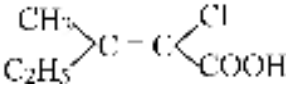
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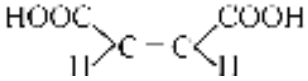
(4) 

45) The 'Z'-isomer is :

(1) 

(2) 

(3) 

(4) 

BIOLOGY

1) Restriction enzymes which cut the DNA a little away from the centre of palindrome site will produce :-

(1) Sticky ends

(2) Blunt ends

(3) Either sticky or blunt ends

(4) Unequal fragments

2) Match the Column-I with Column-II :-

Column - I		**Column - II**	
(i)	Primers	(A)	PCR
(ii)	Separation & purification of products	(B)	Chilled C_2H_5OH
(iii)	Precipitation of DNA	(C)	Gene gun
(iv)	Gene transfer	(D)	Downstream processing

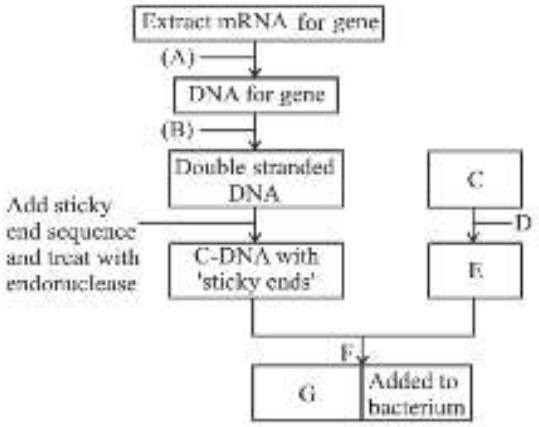
(1) (i) - A, (ii) - D, (iii) - B, (iv) - C

(2) (i) - B, (ii) - A, (iii) - D, (iv) - C

(3) (i) - D, (ii) - A, (iii) - C, (iv) - B

(4) (i) - A, (ii) - B, (iii) - D, (iv) - C

3) Identify the labelled item A, B, C, D, E, F and G in the diagram below from the List-I-VII given



with components :-

(I) DNA polymerase

(II) Plasmid

(III) Plasmid with 'sticky end'

(IV) DNA Ligase

(V) Restriction endonuclease

(VI) Recombinant DNA

(VII) Reverse transcriptase

The correct components are :-

	A	**B**	**C**	**D**	**E**	**F**	**G**
(1)	I	II	IV	VI	III	V	VII
(2)	VII	I	II	V	III	IV	VI
(3)	VII	V	III	I	II	IV	VI
(4)	VII	VI	V	IV	III	II	I

(1) 1

(2) 2

(3) 3

(4) 4

4) Why has the Indian government setup organisation like G.E.A.C. (Genetic engineering Approval Committee) :-

(1) To promote the sale of GM crop in foreign markets.

(2) To make decision regarding the validity of GM research and ensure the safety of introducing GM organism for public use.

(3) To ban all GM research in India .

(4) To manufacture Genetically modified organism for public distribution.

5) Match the Column-A with Column-B :-

	Column-A		**Column-B**
(A)	Bt crop	(1)	Genetically modified to resist insects
(B)	Golden rice	(2)	Enhanced with vitamin A precursor
(C)	RNA interference	(3)	Silence genes to confer pest resistance

(D)	Herbicide tolerant crops	(4)	Resistant to specific weed-killing chemicals

(1) A → (1); B → (2); C → (3); D → (4)

(2) A → (2); B → (1); C → (3); D → (4)

(3) A → (4); B → (1); C → (3); D → (1)

(4) A → (4); B → (3); C → (1); D → (1)

6) Blood cholesterol lowering agent is :-

(1) Cyclosporin-A

(2) Statin

(3) Streptokinase

(4) Ergosterol



(A) (B)

7) Select the correct option for given diagram.

(1) A-Hypopharynx, B-Maxilla

(2) A-Maxilla, B-Mandible

(3) A-Maxilla, B-Labium

(4) A-Hypopharynx, B-Labrum

8) Pick the odd one out based on symmetry:

(1) Hydra

(2) Aurelia

(3) Starfish (adult)

(4) Earthworm

9) In which of following body is covered by dry and cornified epidermal scales or scutes and tympanum represents ear ?

(1) Fish

(2) Amphibian

(3) Reptiles

(4) Aves

10) **Assertion (A):** In Mollusca, mantle cavity plays a role in both respiration and excretion.
Reason (R): Mantle cavity contains feather-like gills used for gaseous exchange and waste removal.

(1) Both A and R are true, and R is the correct explanation of A

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, R is false

(4) A is false, R is true

11) Given below are two statements :-
Statement-I : Members of subphylum vertebrata possess notochord during the embryonic period.
Statement-II : All chordates are vertebrates but not all vertebrates are chordate.
Choose the correct option from the following

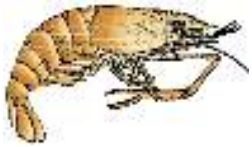
(1) Statement-I is true but statement II is false.

(2) Statement-I is false but statement II is true.

(3) Both statement I and Statement II are true.

(4) Both statement I and statement II are false.

12) Match the Column-I with Column-II and Column-III :-

Column-I		**Column-II**		**Column-III**	
(I)		(A)	Aquatic	(P)	Antenna absent
(II)		(B)	Tracheal system	(Q)	Gregarious pest
(III)		(C)	Locusta	(R)	Crustaceans
(IV)		(D)	Book lungs	(S)	Malpighian tubules

(1) I-B-S, II-D-P, III-A-R, IV-C-Q

(2) I-A-P, II-B-Q, III-C-R, IV-D-S

(3) I-D-S, II-C-R, III-B-Q, IV-A-P

(4) I-B-P, II-C-R, III-D-Q, IV-A-S

13) Which of the following statement **correct** regarding frog.

(1) Ventricle opens into a sac like conus arteriosus on the dorsal side of heart.

(2) Sinus venosus receive deoxy blood from right atrium.

(3) Bidder's canal is present only in male frog kidney.

(4) Copulatory pad is present on first digit of hind limb in male frog.

14) In frog chyme is passed to the ____?

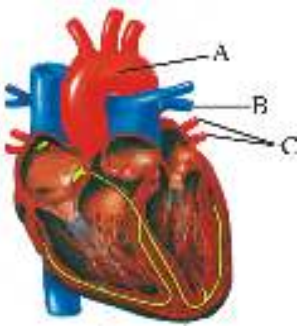
(1) Duodenum
(2) ileum
(3) Jejunum
(4) Cloaca

15) Cell that produce and secrete fibres is -

(1) Adipose cell
(2) Fibroblast
(3) Mast cells
(4) Macrophage

16) ECG depicts the depolarization and repolarization processes during the cardiac cycle. In the ECG of a normal healthy individual, one of the following waves is not represented ?

(1) Depolarization of atria
(2) Repolarization of atria
(3) Depolarization of ventricles
(4) Repolarization of ventricles



17) Find the correct labelling of following diagram.

(1) A-aorta, B-Pulmonary artery,
C-pulmonary veins
(2) A-aorta, B-Pulmonary vein,
C-Left atrium
(3) A-venacava, B-Pulmonary veins,
C-Left atrium
(4) A-Pulmonary artery, B-Systemic arch,
C-pulmonary veins

18)

Select the correct order of events occurring in blood clotting :
(I) Formation of fibrin.
(II) Formation of clot.
(III) Thromboplastin formation.
(IV) Conversion of prothrombin to thrombin.

Choose the correct option :

(1) III, IV, II and I

(2) III, II, I and IV

(3) III, IV, I and II

(4) IV, I, III and II

19) Read the column-I, II and III and identify their correct match -

Column-I		Column-II	Column-III
(1)	Tricuspid valve	Between left atrium and left ventricle	Prevents back flow
(2)	Eosinophils	60-65 percent	Phagocytic cells
(3)	Basophils	0.5-1 percent	Secrete histamine, heparin
(4)	Neutrophils	2-3 percent	Allergy

(1) 1

(2) 2

(3) 3

(4) 4

20) Pulmonary ventilation of air takes place between :-

(1) Atmosphere and alveoli

(2) Tissue and cells

(3) Lungs and pulmonary capillaries

(4) Tissue capillaries and tissue

21) How many cellular layers are present in diffusion membrane of our lungs :-

(1) 2

(2) 3

(3) 4

(4) 1

22) **Statement-I :-** Every 100 ml of oxygenated blood can deliver around 5 ml of O_2 to tissue under normal physiologic conditions.
Statement-II :- When oxygen dissociation curve shift to right means that increase in affinity of Hb with O_2.

(1) Statement-I is true but statement-II is false.

(2) Statement-I is false but statement-II is true.

(3) Statement-I & statement-II both are true.

(4) Both statement-I & statement-II are false.

23) Tubular fluid is ___(A)____ to plasma as it enters Bomman's capsule, ___(B)___ to plasma in PCT, ___(C)___ to plasma at the lower part of loop in medulla, ___(D)___ to plasma as it leaves the LOH to enter the DCT and become ___(E)___ to plasma when leaves the CD (Respectively)

(1) Isotonic, Hyperosmotic, Hyperosmotic, Isotonic, Hypertonic

(2) Isotonic, Isotonic, Hyperosmotic, Hypoosmotic, Hypertonic

(3) Isotonic, Isotonic, Hypoosmotic, Hyperosmotic, Hypotonic

(4) Isotonic, Hypoosmotic, Hypoosmotic, Isotonic, Hypotonic

24) Which of the following statement is incorrect ?

(1) Human lungs eliminated around 200 ml/min of CO_2

(2) Liver is the main site for elimination of cholesterol, bile pigments

(3) Integument in many aquatic animals excretes ammonia in surrounding medium by diffusion.

(4) An adult human excretes on an average 1 to 1.5 gm of urea per day.

25) _____ in the body are activated by changes in blood volume, body fluid volume & ionic concentration.

(1) Chemreceptors

(2) Nociceptors

(3) Osmoreceptors

(4) Thermoreceptors

26) A nerve impulse is transmitted from one neuron to another through junction called as ?

(1) Neuron muscular junction

(2) Synapse

(3) Telodendria

(4) Node of ranvier

27)

Column-I		**Column-II**	
(A)	Multipolar	(i)	Retina of eye
(B)	Bipolar	(ii)	Cerebral Cortex
(C)	Unipolar	(iii)	Cell body
(D)	Nissl's granules	(iv)	Embryonic stage

	A	**B**	**C**	**D**
(1)	ii	iii	i	iv
(2)	ii	i	iii	iv
(3)	ii	iv	i	iii

(4)	ii	i	iv	iii

(1) 1

(2) 2

(3) 3

(4) 4

28) **Statement I :** The neural organisation in Hydra it is composed of a network of neurons. The neural system is better organised in insects, where a brain is present along with a single nerve cord and neural tissues. The vertebrates have a more developed neural system.
Statement II : The neural system of all animals is composed of highly specialised cells called neurons which can detect, receive and transmit different kinds of stimuli.

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

29) **Assertion (A) :** Posterior pituitary is also called neurohypophysis.
Reason (R) : It stores and releases hormones synthesized by the hypothalamus.

(1) Both A & R correct and R is the correct explanation of A.

(2) Both A & R correct and R is the not correct explanation of A.

(3) A is correct but R is incorrect.

(4) A is incorrect but R is correct.

30) Which of the following statement is incorrect for thyroid gland ?

(1) The thyroid gland is composed of two lobes which are located on either side of trachea.

(2) Cancer of thyroid gland or development of nodules of thyroid gland causes hypothyroidism

(3) The thyroid gland is composed of follicles and stromal tissue.

(4) Follicular cells synthesis two hormone t_4 and t_3.

31) Male sexual behaviour is influenced by :-

(1) ACTH

(2) Oestrogen

(3) Testosterone

(4) Progesterone

32) Read the following statements -
(A) Hormones are non-nutrient chemicals which act as intercellular messengers and are produced in large amounts.
(B) Adrenal, pancreases, parathyroid and thymus are the organised exocrine glands in our body.
(C) Invertebrates possess very simple endocrine systems with few hormones.
(D) Endocrine glands lack ducts and are hence, called ductless glands.
How many of the above statements are correct?

(1) Three

(2) Four

(3) Only One

(4) Two

33) __Q__ is an auto immune disorder affecting neuromuscular junction leading to fatigue weakening and paralysis of skeletal muscle.

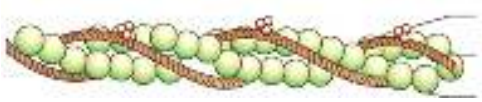
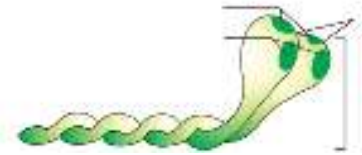
(1) Q = Tetany

(2) Q = Osteoporosis

(3) Q = Myasthenia gravis

(4) Q = Muscular dystrophy

34) Identify the following structure related with muscle shown below and select the correct. Option for together.

	 (a)	 (b)
(1)	Thin Filament	Actin
(2)	Thick filament	Thin filament
(3)	Thin filament	Meromyosin
(4)	Light Meromyosin	Heavy Meromyosin

(1) 1

(2) 2

(3) 3

(4) 4

35) The Human cranium is made up of how many bones.

(1) 14

(2) 8

(3) 10

(4) 12

36) **Assertion :** In testis, the immature male germ cells (spermatogonia) produce sperms spermatogenesis.
Reason : A primary spermatocyte completes the first meiotic division (Reduction division) leading to formation of two equal haploid cell called secondary spermatocyte, which have only 23 chromosomes

each.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is correct but **Reason** is incorrect.

(4) **Assertion** is incorrect but **Reason** is correct.

37) **Statement A:** During parturition, continuation of stimulatory reflex between the uterine contraction and oxytocin secretion results in stronger and stronger contractions.
Statement B: The mammary glands start producing milk towards the end of pregnancy by the process called lactation.

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

38) Which of the following represents the correct sequence of structures through which sperm pass from the testes to the external urethral opening?

(1) Seminiferous tubules → Vasa efferentia → Rete testis → Epididymis → Vas deferens → Ejaculatory duct → Urethra → Urethral meatus

(2) Seminiferous tubules → Epididymis → Rete testis → Vasa efferentia → Vas deferens → Ejaculatory duct → Urethra → Urethral meatus

(3) Rete testis → Seminiferous tubules → Vasa efferentia → Epididymis → Vas deferens → Urethra → Ejaculatory duct → Urethral meatus

(4) Seminiferous tubules → Rete testis → Vasa efferentia → Epididymis → Vas deferens → Ejaculatory duct → Urethra → Urethral meatus

39) **Assertion (A) :** Vasectomy prevents spermatogenesis.
Reason (R) : Vasectomy involves cutting vas deferens, so sperm cannot reach ejaculatory duct.

(1) Both A and R true, R is correct explanation

(2) Both A and R true, R not correct explanation

(3) A true, R false

(4) A false, R true

40) Which one is a copper-releasing IUD?

(1) Lippes loop

(2) Multiload 375

(3) LNG-20

(4) Vaginal ring

41) **Statement-I :** Amniocentesis procedure is performed during pregnancy.
Statement-II : In amnio centesis some of the amiotic fluid of the developing socity is taken to analyze the fetal cell.

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Statement-I is incorrect but statement-II is correct

42) Find out the incorrect statement regarding cancer :-

(1) Computed tomography uses X-rays to generate a three-dimensional image of the internals of an object.

(2) Techniques like radiography, CT and MRI are very useful to detect cancers of the internal organs.

(3) Matastasis is the most feared property of malignant tumor.

(4) Carcinogens are agents that prevent cancer growth.

43) Match the following medicinal plants with their products :-

	Column-I (Plant)		**Column-II (Product)**
A.	Papaver somniferum	1	Cocaine
B.	Catharanthus roseus	2.	Morphine
C.	Erythroxylum Coca	3.	Vinblastine
D.	Cannabis Sativa	4.	Hasish

(1) A-2, B-3, C-1, D-4

(2) A-3, B-2, C-4, D-1

(3) A-1, B-3, C-2, D-4

(4) A-4, B-1, C-3, D-2

44) Read the following statements :-
A. In ascariasis, a healthy person acquires infection through contaminated water, vegetables and fruits etc.
B. *Entamoeba histolytica* is a protozoan parasite in small intestine of human.
C. Malignant malaria is caused by *plasmodium vivax*.
D. Typhoid fever could be confirmed by widal test
E. Plasmodium enters the human body as sporozoites
Select the option which include all the correct statements only :-

(1) A, B, C

(2) B, C, D, E

(3) A, C, D, E

(4) A, D, E

45) Genetic drift can be defined as :-

(1) Systematic change of gene frequencies.

(2) Random change of gene/allelic frequencies

(3) Systematic change in habitat

(4) New combination of genes during gametogenesis

46) Consider the following four statements (a - d) and select the option which includes all the correct ones only

(a) Slime moulds are also called naked fungi

(b) Bacteriophage generally contain double stranded DNA

(c) Potato spindle tuber disease is caused by prions

(d) Male cone is absent in *Cycas* plant

(1) Statements (a), (b) and (d) correct

(2) Statements (a) and (b) correct

(3) Statements (a), (b), (c) are correct

(4) Statements (a), (b), (c), (d) correct

47)

Identify A, B, C and D in given table :-

Name of organism	**Habitat**	**Chlorophyll Pigment**	**Store food**
Red algae	A	'a' &'d'	Floridian starch
Brown algae	Marine water	B	Laminarian & Mannitol
Dino-flagellates	Marine water	C	Starch
Green algae	D	'a' &'b'	Starch

	A	**B**	**C**	**D**
(1)	Marine water	a,b	a,d	Tree
(2)	Salty water	a,d	a,b	Animal body
(3)	Mostly fresh water	a,c	a,c	Land
(4)	Mostly marine water	a,c	a,c	Water

(1) 1

(2) 2

(3) 3

(4) 4

48) **Assertion:** Genomic DNA of *E. coli* is considered naked.
Reason : Prokaryotes are devoid of nuclear envelope.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

49) **Assertion :** Ascomycetes fungi are parasitic, saprophytic or coprophilous.
Reason : Mycelium of ascomycetes fungi is coenocytic.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

50) Which of the following is not a feature of Bentham and Hooker's classification ?

(1) It was based on gross morphology
(2) It was a natural classification
(3) It was related to seeded plants
(4) It was a phylogenetic classification

51) Which of the following statement is correct regarding *Agaricus* and *Aspergillus*?

(1) They produce ascus and basidium respectively
(2) They are producers and decomposers respectively
(3) They produce asexual spores and gametes respectively
(4) They are member of club fungi and sac fungi respectively.

52) According to five kingdom classification of Whittakar the virus, viroids and lichens are placed in which kingdom?

(1) Monera
(2) Protista
(3) Fungi
(4) Not in any kingdom

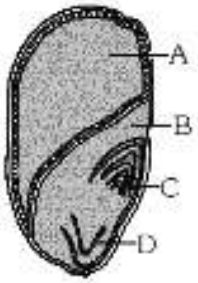
53) Cells of collenchgyma have thickend corners due to the deposition of

(1) Lignin
(2) Subrin
(3) Cutin
(4) Pectin

54) Radial conduction of water takes place by :-

(1) Ray parenchyma
(2) Tracheids
(3) Vessels
(4) Fibres

55) Identify A, B, C and D in the given figure showing L.S. of a monocot seed and select the correct



option.

	A	B	C	D
(1)	Endosperm	Scutellum	Plumule	Radicle
(2)	Endosperm	Scutellum	Radicle	Plumule
(3)	Scutellum	Endosperm	Radicle	Plumule
(4)	Plumule	Radicle	Scutellum	Endosperm

(1) 1
(2) 2
(3) 3
(4) 4

56) Part of chromosome which is related with the ageing process in human:

(1) Chromatid
(2) Satellite
(3) Kinetochore
(4) Telomere

57) From the following select the statement that is true:-

(1) All cells have a cell wall
(2) Animal cells contain microtubules but plant cells do not contain microtubules
(3) The golgi apparatus is found only in animal cells
(4) Chloroplasts are found in plant cells but not in prokaryotic or animal cells

58) Significance of meiosis lies in-

(1) production of genetic variability
(2) maintaining the number of chromosomes through generations
(3) forming the basis of evolution

(4) All of the above

59) Vinblastin, Gum and Rubber all are :-

(1) Secondary metabolite
(2) Primary metabolite
(3) Protein
(4) Lipid

60) Which element is present in more quantity in human body ?

(1) Si
(2) O
(3) N
(4) Mg

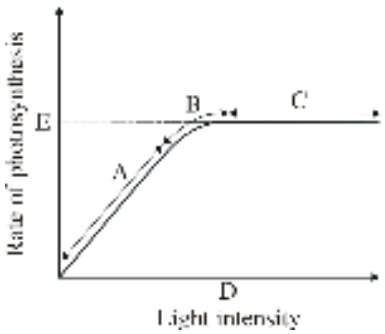
61) Which of the following is incorrectly matched pair ?

(1) Ethylene – Cousins
(2) Kinetin – Skoog and Miller
(3) Gibberellic acid – F.W. Went
(4) Auxin – Charles Darwin and Francis Darwin

62) **Statement-I :** Plants follow different pathways in response to environment or phases of life to form different kinds of structures, this ability is called plasticity.
Statement-II : Heterophylly in cotton, coriander and larkspur is an examples of plasticity.

(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Statement-I & Statement-II both are incorrect.

63)



In diagram A, B, C and D represents.

(1) A → Light is limiting factor, B → Optimum, C → Other factors except light became limiting, D → Saturation

(2) A → Other factors except light become limiting,
B → Optimum, C → Saturation,
D → Light is limiting factor

(3) A → Saturation, B → Optimum,
C → Light is a limiting factor,
D → Other factors except light became limiting

(4) A → Light is limiting factor, B → Saturation,
C → Other factors except light became limiting,
D → Optimum

64) Photochemical reaction takes place in :-

(1) Grana

(2) Stroma

(3) Cristae

(4) Matrix

65) The complete oxidation of pyruvate by the stepwise removal of all the hydrogen atoms, leaving three molecules of CO_2. This includes -

(1) Link reaction only

(2) Kerb's cycle only

(3) Both 1 and 2

(4) ETS only

66) Which of the following is incorrect statement?

(1) The conversion of 1, 3 - BPGA to 3-phosphoglyceric acid (PGA), is an energy yielding process.

(2) Two redox-equivalents are removed (in form of two hydrogen atom) from PGAL and transferred to a molecule of NAD^+

(3) PGAL is reduced and with inorganic phosphate to get converted into BPGA.

(4) ATP is synthesized during the conversion of PEP to pyruvic acid.

67) Find out the correct match from following table :-

	Characteristics	**C_3 plants**	**C_4 plants**
I	Primary CO_2 acceptor	RuBP	RuBP
II	Primary CO_2 fixation product during C_3 cycle	PGA	PGA
III	No. of carbons in the primary CO_2 acceptor	3	5

IV	Initial carboxylation reaction occurs	Mesophyll cells	Bundle sheath cells

(1) I, III, IV

(2) II only

(3) IV only

(4) II, III, IV

68) Which of the following statement is correct about apomixis ?

(1) Apomixis is a type of sexual reproduction

(2) Apomixis can help to fix hybrid vigour across generations.

(3) Apomixis leads to the formation of genetically variable seeds.

(4) Apomixis involves meiosis and fertilisation.

69) **Assertion (A) :** Synergids are two specialised cells adjacent to the egg cell in the embryo sac that play a crucial role in guiding the pollen tube towards the egg for successful fertilisation.
Reason (R) : Both synergids degenerate before the pollen tube enters the embryo sac.

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**

(3) **(A)** is true but **(R)** is false

(4) **(A)** is false but **(R)** is true

70) How many gametes are involved in double fertilization.

(1) 2

(2) 3

(3) 4

(4) 5

71) When a single gene can exhibit multiple phenotypic expression such a gene represents :

(1) Polygenic inheritance

(2) Pleiotropy

(3) Multiple allele

(4) Incomplete dominance

72) **Assertion (A) :** Homozygous (Hb^S Hb^S) individuals appear affected under low oxygen concentration.
Reason (R) : RBCs undergoes polymerisation it's shape under low oxygen tension causing change in.

(1) Both A and R true , Reason is correct explanation of Assertion.

(2) Both A and R true, Reason is not correct explanation of Assertion

(3) A is true and R is false

(4) A is false and R is true

73) In F_2 generation of Mendelian dihybrid cross, 2000 plants are obtained. How many of them have atleast one recessive trait or phenotype ?

(1) 1125
(2) 875
(3) 1000
(4) 1250

74) **Statement-1 :** When F_1 plant of Snapdragon was pollinated then F_2 genotype ratio was exactly as we would expect in any mendelian monohybrid cross.
Statement-2 : Starch is synthesized effectively by BB homozygotes and in contrast, bb homozygotes have lesser efficiency in starch synthesis. Heterozygotes produce intermediate seeds, then from this angle, the alleles show dominance.

(1) Both statements 1 and 2 are correct
(2) Both statements 1 and 2 are incorrect
(3) Statement-1 is correct and statement-2 is incorrect
(4) Statement-1 is incorrect and statement-2 is correct

75) Which of the following is *incorrect* for garden pea?

	Character	**Genotype**	**Phenotype**
(1)	Flower position	Homozygous or heterozygous	Terminal
(2)	Seed colour	Homozygous or heterozygous	Yellow
(3)	Pod colour	Homozygous or heterozygous	Green
(4)	Pod shape	Homozygous or heterozygous	Inflated

(1) 1
(2) 2
(3) 3
(4) 4

76) Fill in the blanks :
The VNTR belongs to a class of satellite DNA referred a as _____. Size of VNTR varies in size from _____.

(1) Micro satellite, 0.1 to 20 kb
(2) Mini satellite, 0.1 to 20 kb
(3) Micro satellite, 1 to 20 kb
(4) Mini satellite, 1 to 20 kb

77) **Assertion (A) :-** Regulation of lac operon by repressor is referred to as negative regulation.
Reason (R) :- The repressor protein binds to the operator region of the operon and prevents RNA

polymerase from transcribing the operon.

(1) Assertion is true but reason is false

(2) Both A & R are false

(3) A & R both are true but R is not correct explanation of A

(4) A & R both are true and R is correct explanation of A

78) **Assertion (A):** Splicing is necessary in eukaryotic cells for producing functional mRNA.
Reason (R): In eukaryotes, introns are coding sequences and must be retained in mature RNA.

(1) Both A and R are true, and R is the correct explanation of A

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, R is false

(4) A is false, R is true

79) Match the following with respect to Transcription in Prokaryotes:

	Column I		**Column II**
1.	σ (Sigma) factor	(a)	Helps in promoter recognition and initiation
2.	Template strand	(b)	Serves as actual template for RNA synthesis
3.	Coding strand	(c)	Same sequence as RNA (except T→U)
4.	RNA Polymerase	(d)	Catalyzes formation of RNA

(1) 1-a, 2-b, 3-c, 4-d

(2) 1-d, 2-b, 3-c, 4-a

(3) 1-d, 2-c, 3-a, 4-b

(4) 1-c, 2-d, 3-a, 4-b

80) If the sequence of base in coding strand of DNA is ATTCGATG, then the sequence of base in mRNA will be:

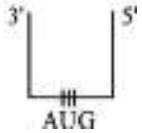
(1) TAAGCTAC

(2) UAAGCUAC

(3) ATTCGATG

(4) AUUCGAUG

81) Which of the following tRNA doesn't exist:

(1) 

(2) 

(3) 

(4) 

82) A molecular biologist analysed a nucleic acid sample and found that it contained the N_2 bases in following proportion A = 20%, T = 15%, C = 35%, G = 30% this data indicates that :-

(1) Nucleic acid in the sample is single stranded DNA
(2) Nucleic acid in the sample is double stranded DNA
(3) Nucleic acid in the sample is double stranded RNA
(4) Nucleic acid in the sample is single stranded RNA

83) From the following, identify the correct combination of salient features of Genetic Code :-

(1) Universal, Non-ambiguous, Overlapping
(2) Degenerate, Overlapping, Commaless
(3) Universal, Ambiguous, Degenerate
(4) Degenerate, Non-overlapping, Non- ambiguous

84) Given below are two statement :-
Statement-I : Mycorrhizae are association between fungi and algae.
Statement-II : The mediterranean orchid Ophrys employs 'sexual decit' to get pollination done by a species of bee.

(1) Both statement-I and statement-II are correct.
(2) Both statement-I and statement-II are incorrect.
(3) Statement-I is correct and statement-II is incorrect.
(4) Statement-I is incorrect and statement-II is correct.

85) Given below are two statements :-
Statement-I : Predators also help in maintaining species diversity in a community.
Statement-II : Besides acting as conduits for energy transfer across trophic level, predators play

other important roles.
In the light of above statements, choose the correct answer from given below options ?

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

86) Given below are two statements :-
Statement-I : Organisms at each trophic level depends on those at the lower trophic for their energy demands.
Statement-II : The number of trophic level in the detritus food chain is restricted as the transfer of energy follows 10 percent law.
In the light of the above statements, choose the most appropriate answer from the option given below :-

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

87) **Assertion (A) :** Plants, animals and micro-organism together constitute biotic components.
Reason (B) : Rocks, soil, air, water, light and temperature are the abiotic components of an environment.

(1) Both (A) and (R) are true and (R) is correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

88) Read the following statement and select correct one :-

(1) Gross primary productivity is the available biomass for the consumption to heterotrophs,

(2) Secondary productivity depends on the photosynthetic capacity of the plant species inhabiting a particular area.

(3) Productivity is expressed in terms of $Kg^{-2}\,yr^{-1}$ or $(KCalm^{-2})\,yr^{-1}$ to compare the productivity of different ecosystems.

(4) The annual NPP of the whole biosphere is 170 billion tons of organic matter.

89) Which of the following statements is incorrect ?

(1) Sacred groves are examples of in-situ conservation.

(2) Seed banks are ex-situ conservation.

(3) Zoological parks represent in-situ conservation.

(4) Cryopreservation is a ex-situ technique.

90) Given below are two statements:

Statement-I: Speciation is generally a function of time, unlike temperate regions subjected to frequent glaciations in the past, tropical latitudes have remained relatively undisturbed for millions of years and thus, had a long evolutionary time for species diversification.
Statement-II: Tropical environments, unlike temperate ones, are less seasonal, relatively more constant and predictable. Such constant environments promote niche specialisation and lead to a greater species diversity.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.